

Lecture 29 — Same mathematics, very different computational cost

Fri Oct 30

Last updated: October 8, 2026.

Reading: Boyd's $ab^T c$ example; [micro-lab](#). Related objectives: [Lectures](#). [Schedule](#).

Note: Precursor to the capstone — same mathematical answer, different storage and runtime depending on order.

Bring paper. Short lecture (warm-up $\rightarrow$ fill-in $\rightarrow$ board $\rightarrow$ Your turn), then the [micro-lab](#) same day.

Learning objectives

By the end of class, students should be able to:

1. Compare algebraically equivalent orders of computation.
2. Count intermediate values that must be stored.
3. Measure runtime for increasing problem sizes.
4. Explain why forming a large intermediate matrix may be wasteful.
5. Choose a computational order and justify it quantitatively.

Warm-up

Let $a = (1, 2)$, $b = (3, 4)$, $c = (1, -1)$ (column vectors).

- (a) Form the outer product ab^T (a 2×2 matrix).
- (b) Compute $(ab^T)c$ and $a(b^T c)$. Same vector?

Fill in

Associativity. For compatible sizes, $(XY)z = X(Yz)$. In particular, for n -vectors a, b, c ,

$$(ab^T)c = a(b^T c).$$

Same mathematical result — different _____.

Order 1 — form the matrix first. Compute $M = ab^T$ ($n \times n$), then Mc .

- Intermediate storage: about _____ numbers (the whole matrix M).
- Cost idea: building M and multiplying by c scales like _____ work (order n^2).

Order 2 — scalar first. Compute $s = b^T c$ (one number), then sa .

- Intermediate storage: the scalar s , plus the vectors — order _____ numbers, not n^2 .
- Cost idea: two inner/scale passes — order _____ work.

Why wasteful? Forming $ab^\top$ materializes a dense $n \times n$ matrix you may never need as an object; then you immediately collapse it with c . Capstone theme: choose an algorithm that matches the structure, not only the formula on the page.

Runtime. In the micro-lab you will time both orders for growing n and justify which order to use with measured times (and storage counts).

Worked in class

W1. Algebra. With $a = (2, 0, -1)$, $b = (1, 1, 1)$, $c = (1, 0, 1)$, compute $s = b^\top c$ and $a(b^\top c)$. Then form $ab^\top$ and $(ab^\top)c$. Match?

W2. Storage count. For general n , how many entries does $ab^\top$ have? How many numbers do you store for the scalar-first method beyond the three input vectors?

W3. Choose an order. You need $y = (ab^\top)c$ for $n = 5000$ once. Which order do you run, and what two quantitative reasons (storage + flops/runtime idea) support the choice?

W4 (preview). Sketch what a timing table should look like: rows $n = 200, 1000, 3000$; columns “form ab^T then multiply” vs “scalar first.” What qualitative pattern do you expect?

Your turn

Y1. Let $a = (1, -1, 0)$, $b = (2, 1, 3)$, $c = (0, 1, 1)$. Compute both $(ab^T)c$ and $a(b^Tc)$.

Y2. For $n = 100$, about how many intermediate entries does Order 1 store in ab^T ? Order 2 in the scalar s ?

Y3. One sentence: why can two algebraically equal expressions have wildly different runtimes?

Micro-lab, same day

After this short lecture: [Micro-lab — Computational cost](#). Compare the two orders in Octave or Python, count storage, time growing n , and justify a choice on the one-pager.

Answers (Your turn)

Y1. $b^\top c = 2 \cdot 0 + 1 \cdot 1 + 3 \cdot 1 = 4$, so $a(b^\top c) = 4a = (4, -4, 0)$. The matrix route gives the same:

$$ab^\top = \begin{bmatrix} 2 & 1 & 3 \\ -2 & -1 & -3 \\ 0 & 0 & 0 \end{bmatrix}, \quad (ab^\top)c = (4, -4, 0).$$

Y2. Order 1 stores $n^2 = 10,000$ entries in $ab^\top$. Order 2 stores one scalar s as the essential intermediate (beyond the inputs).

Y3. Associativity preserves the value but not the work: different parenthesizations create different intermediates (e.g. an $n \times n$ matrix vs a scalar), so flops and memory — and therefore runtime — can differ by orders of magnitude.